

Theory of Automata

Assignment No: 1

Due on: 15 Sept,2025 (9:00 am)

Submit Hard form in office, and also upload hardform. Otherwise your assignment will not be accepted.

All language are defined over $\Sigma = \{a, b\}$

Question No:1

For the following languages

- a. Write an RE.
- b. Design an FA, and
- c. Design a TG. (only if the FA in part b can be expressed in a compressed way using a TG).
- i. All strings that contain at least one **aba** as a substring.
- ii. All strings that start and end with the same symbol.
- iii. All strings in which the total number of **bs** is even.
- iv. All strings that do not contain the substring **bab**.
- v. All strings that begin with **aa** and contain at least two **bbs**.
- vi. All strings in which every **a** is immediately followed by at least one
- vii. All strings that contain an equal number of **as** and **bs** .
- viii. All strings ending with either **aaa** or **bbb**.
- ix. All strings that never contain **abab** as a substring.
- x. All strings with at least three **as** and at least two **bs**.

Question No:2

Describe (in English) the languages associated with the following regular expressions.

- i. $(a+b)^*aa(a+b)^*$
- ii. $(ab+ba)^*$
- iii. $a^*b^*a^*$
- iv. $(a+bb)^*$
- v. $b^*a(b^*ab^*a^*)^*$

Question No:3

Show that

- I. $((a+bb)^*aa)^*$ is equivalent to $\Lambda + (a+bb)^*aa$ in a sense they define the same language.
- II. $a(ba + a)^*b$ is equivalent to $aa^*b(aa^*b)^*$ In a sense they define the same language.

Question No: 4

- I. Provide a recursive definition of language having all strings with length multiple of 2.
- II. Provide recursive definition of odd palindrome.
- III. Give a recursive definition of the language the language ODDSTRING, i.e. of all words of odd length.

Instructions:

Submit assignment in hard form. Plagiarism gets minimum zero and maximum of F in course. No extension in deadline shall be given.